DO NOW — RECALL

Final specification and synoptic checkpoint

Name: _____ Class: _____ Date: _____

1. Name a PageRank property beyond counting inbound links.

2. What is a page fault?

3. What does an INNER JOIN return?

4. Search value 11 in [1,3,5,7,9,11,13], showing mid indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Weighted contributions from links, damping and iterative calculation.

2. Model answer

Required virtual memory page not resident in RAM.

3. Model answer

Rows matching the join condition.

4. Model answer

mid3 (7) then mid5 (11) found; index5.

By the end, you can…

• Use evidence rather than ticks alone

• Find links between papers

• Prioritise gaps

Vocabulary — know these terms

synoptic — Combining knowledge across topics.

retrieval — Reproduce knowledge from memory.

misconception — Incorrect conceptual understanding.

intervention — Specific teaching targeting a gap.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check reasoning

Set a small test with a target below the midpoint to check the high-bound update independently.

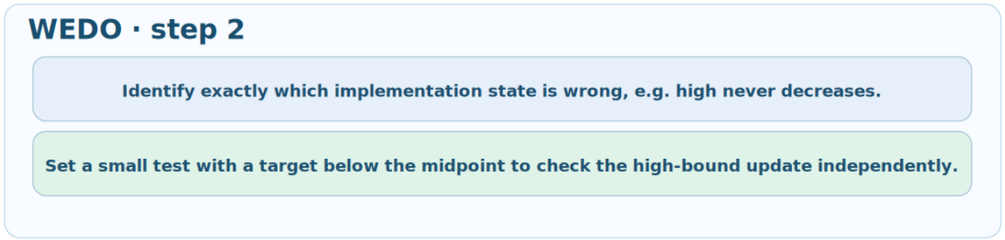
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Identify a paper-crossing weakness: a candidate knows binary search but cannot implement it.

Identify exactly which implementation state is wrong, e.g. high never decreases.



WEDO · step 2: Identify exactly which implementation state is wrong, e.g. high never decreases.; Set a small test with a target below the midpoint to check the high-bound update independently.

NEW LEARNING

Read, annotate and explain

Use evidence rather than ticks alone

A topic is secure when a learner can solve an unfamiliar example, not merely recognise its name.

Find links between papers

Paper 1 numerical representation and database knowledge can underpin Paper 2 programming and algorithm questions.

Prioritise gaps

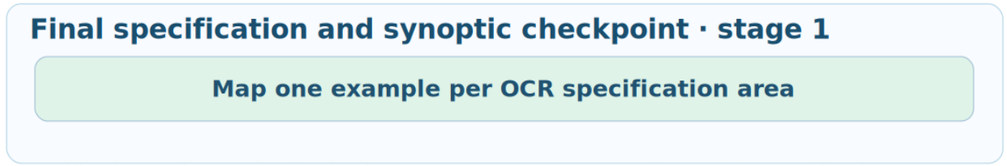
Record the exact misconception or missing step so later Paper 1/2 reteaching sessions target observable weaknesses.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

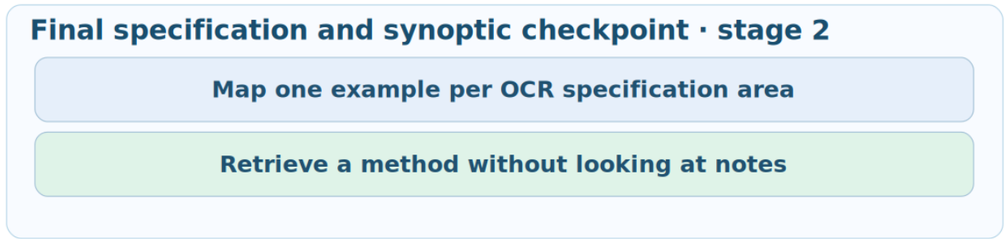
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



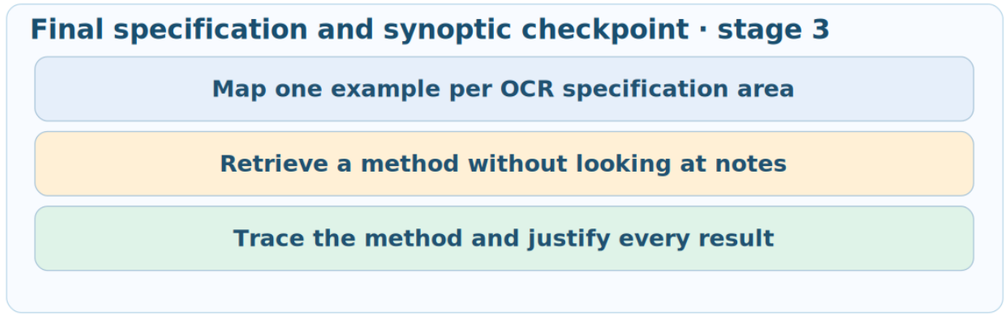
Final specification and synoptic checkpoint · stage 1: Map one example per OCR specification area

New Learning



Final specification and synoptic checkpoint · stage 2: Map one example per OCR specification area; Retrieve a method without looking at notes

New Learning



Final specification and synoptic checkpoint · stage 3: Map one example per OCR specification area; Retrieve a method without looking at notes; Trace the method and justify every result

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Knowing a topic name alone proves that you can apply the associated algorithm.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

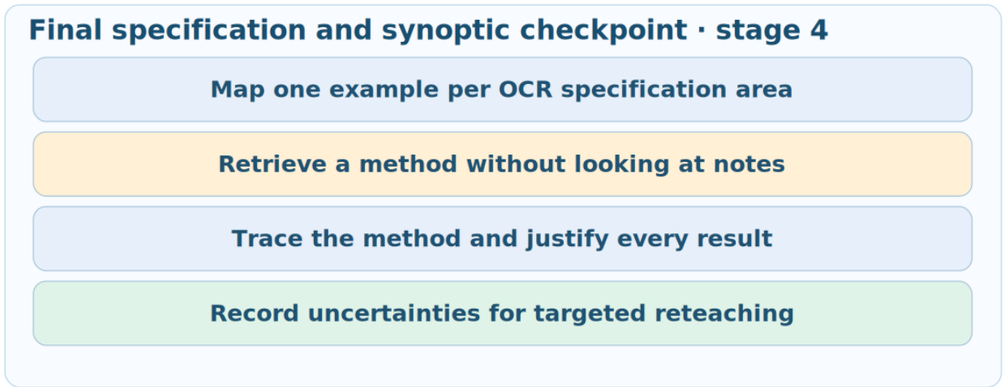
Question 3

A school chooses AI scoring for pupil support. Give three contextual safeguards and explain risks. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Final specification and synoptic checkpoint · stage 4: Map one example per OCR specification area; Retrieve a method without looking at notes; Trace the method and justify every result; Record uncertainties for targeted reteaching

I DO — WORKED EXAMPLE

Follow the worked method

Evaluate the claim 'all sorts take O(n log n) in the worst case'.

Predict first

Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

Check reasoning

Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State roles of PC, MAR, MDR, CIR and ACC, plus one bus role. [6]

Question 2

A sorted list [2,4,6,8,10,12,14] is searched for 12. Trace binary search, then explain worst-case Big O. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Identify exactly which implementation state is wrong, e.g. high never decreases.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Evaluate the claim 'all sorts take O(n log n) in the worst case'.

Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.



IDO · step 1: Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

Evaluate the claim 'all sorts take O(n log n) in the worst case'.

Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.



IDO · step 2: Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.; Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Identify a paper-crossing weakness: a candidate knows binary search but cannot implement it.

Identify exactly which implementation state is wrong, e.g. high never decreases.



WEDO · step 1: Identify exactly which implementation state is wrong, e.g. high never decreases.